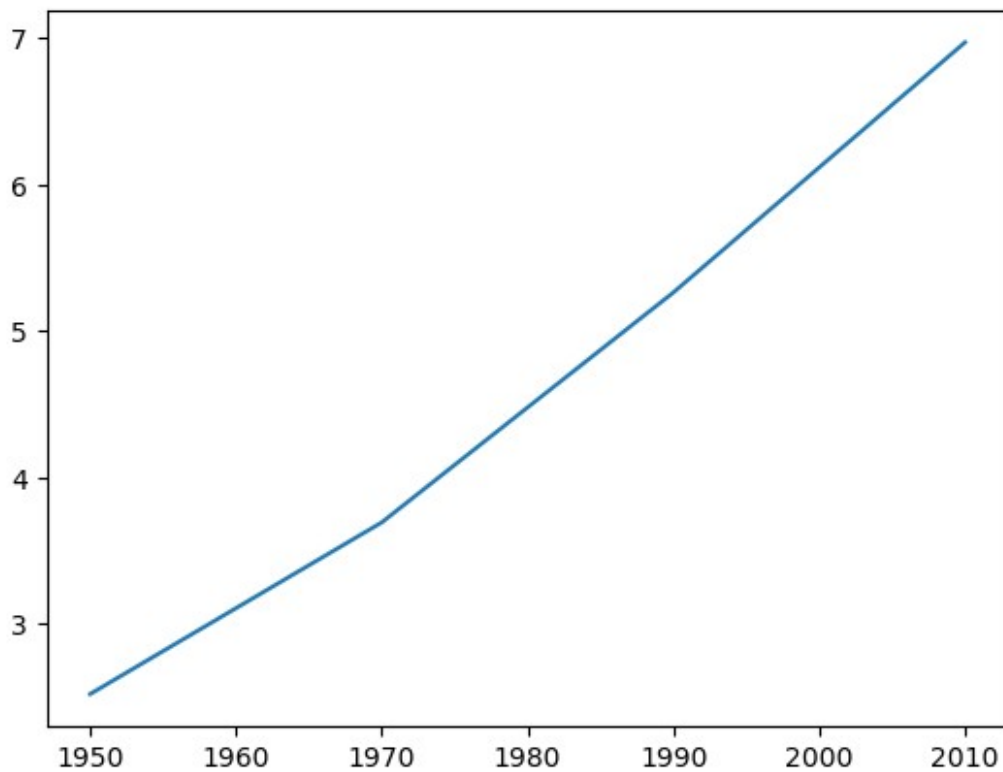


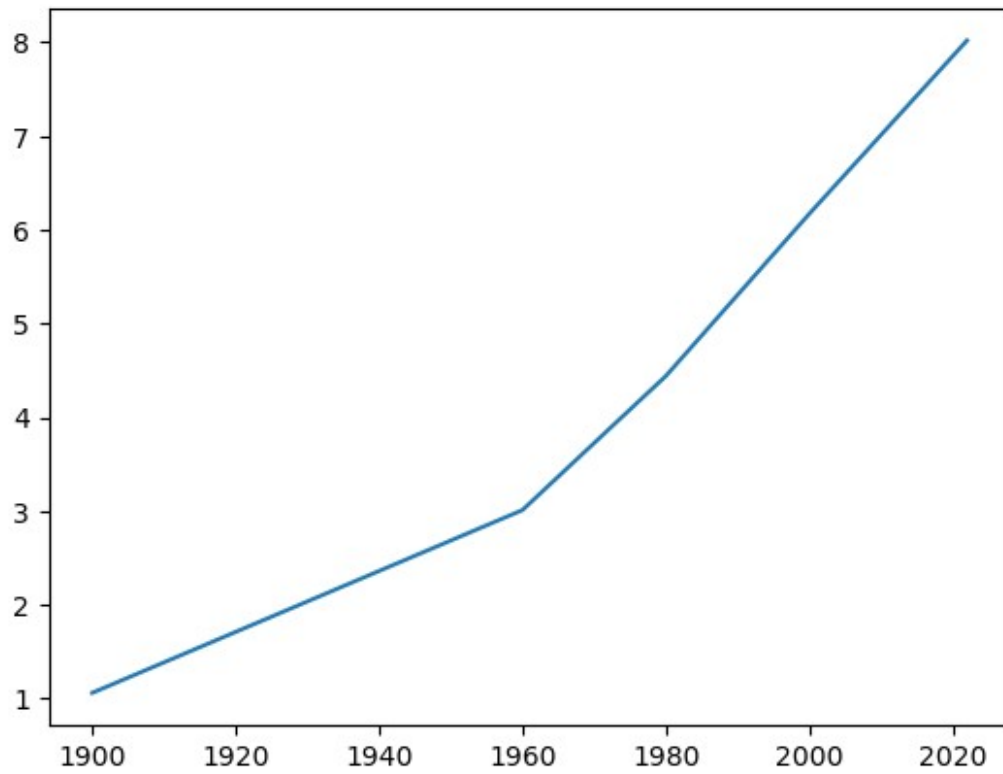
```
#mariem_benali_SSR
```

```
import matplotlib.pyplot as plt  
year = [1950, 1970, 1990, 2010]  
pop = [2.519, 3.692, 5.263, 6.972]  
plt.plot(year, pop)  
plt.show()
```

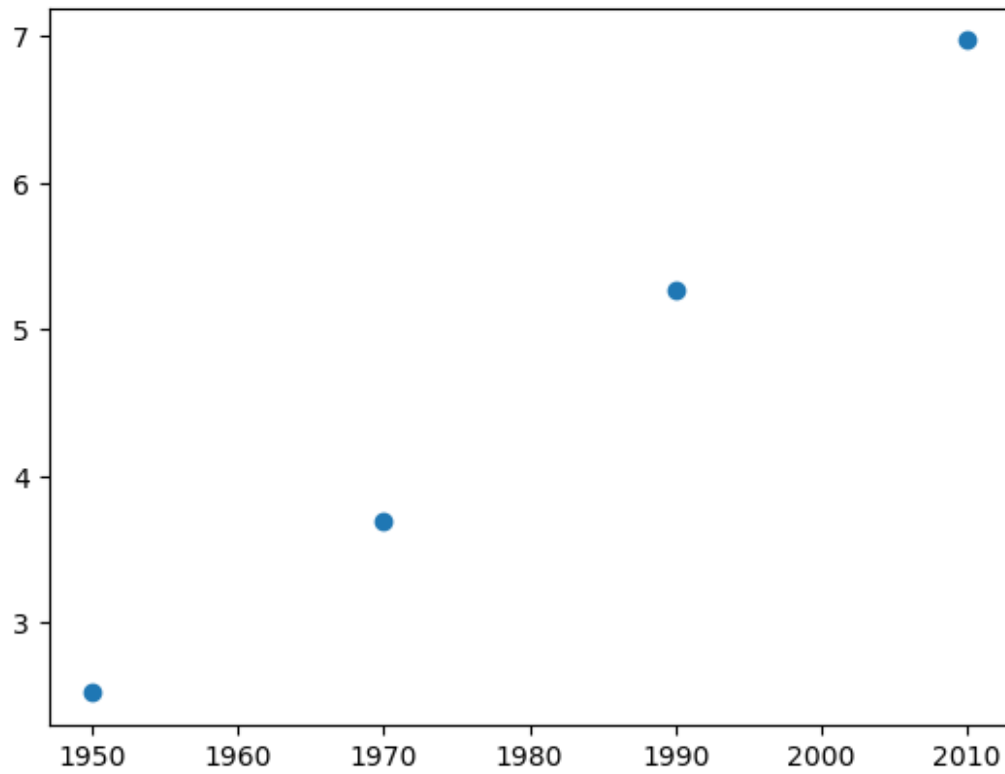


```
#exemple_1_Matplotlib
```

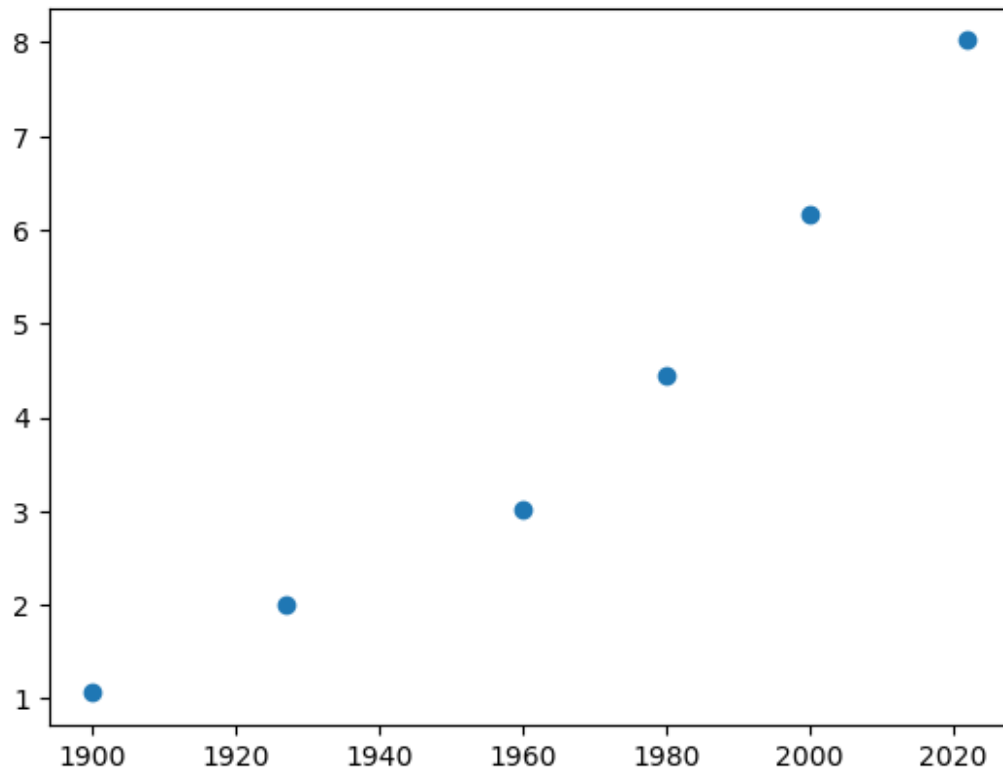
```
import matplotlib.pyplot as plt  
year = [1900, 1960, 1980, 2000, 2022]  
pop = [1.06, 3.01, 4.44, 6.17, 8.02]  
plt.plot(year, pop)  
plt.show()
```



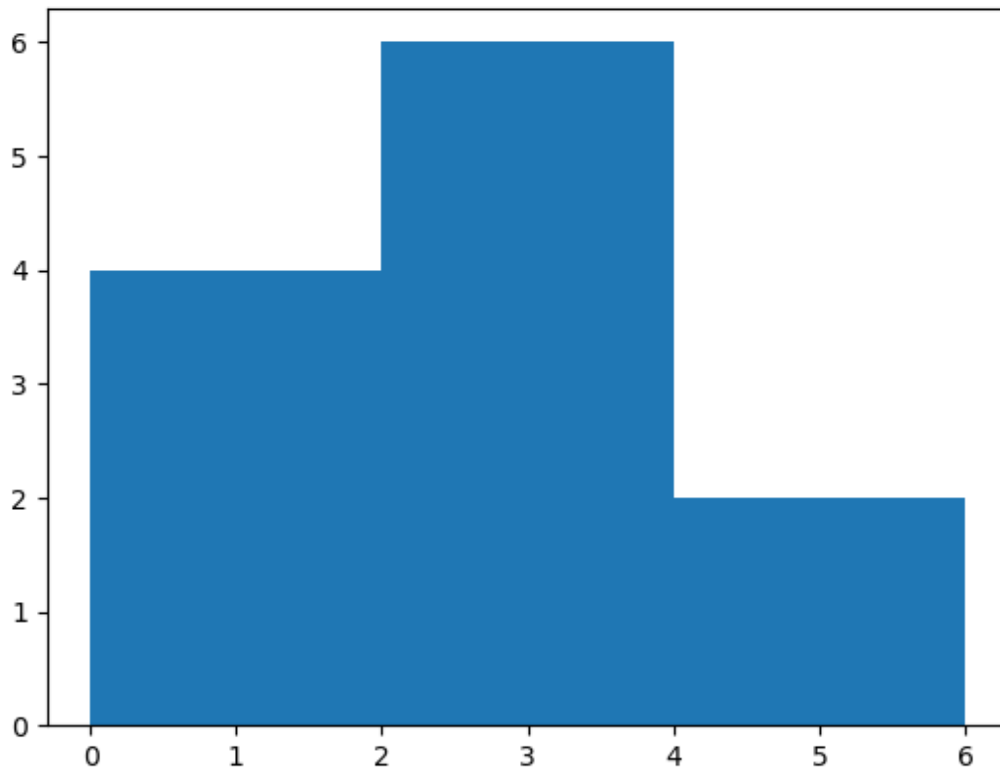
```
import matplotlib.pyplot as plt
year = [1950, 1970, 1990, 2010]
pop = [2.519, 3.692, 5.263, 6.972]
plt.scatter(year, pop)
plt.show()
```



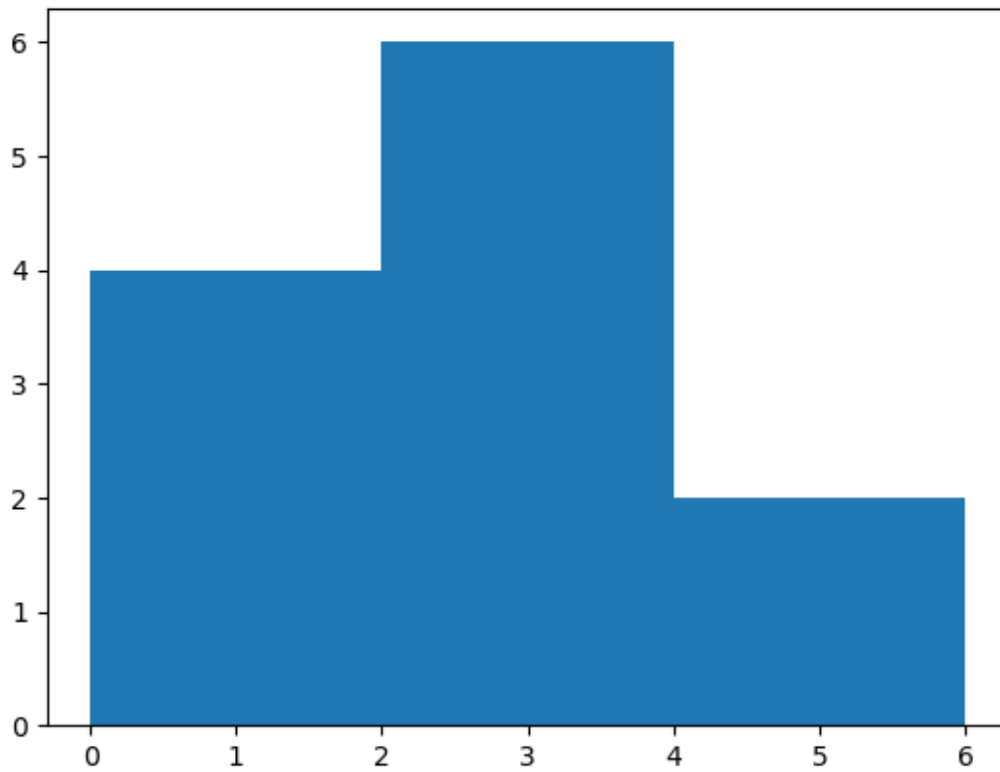
```
#exemple_Scatter plot  
import matplotlib.pyplot as plt  
year = [1900, 1927, 1960, 1980, 2000, 2022]  
pop = [1.06, 2.00, 3.01, 4.44, 6.17, 8.02]  
plt.scatter(year, pop)  
plt.show()
```



```
import matplotlib.pyplot as plt
values = [0,0.6,1.4,1.6,2.2,2.5,2.6,3.2,3.5,3.9,4.2,6]
plt.hist(values, bins=3)
plt.show()
```

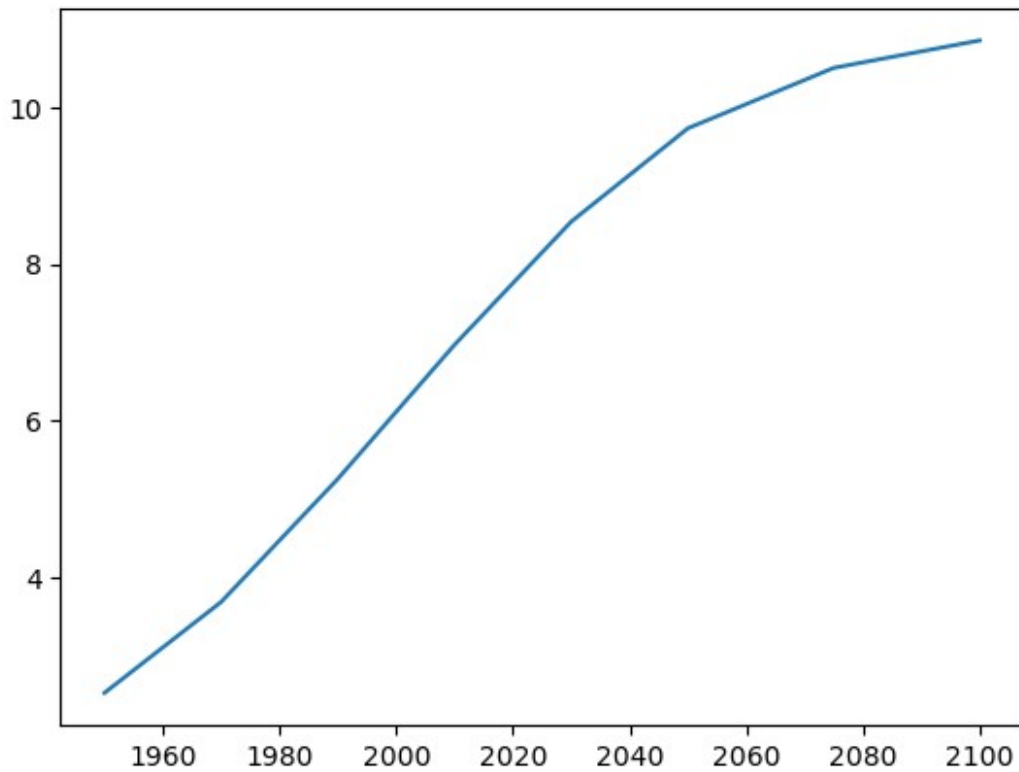


```
#exemple3_Histogram example
import matplotlib.pyplot as plt
values = [0,0.9,1.7,1.9,2.6,2.9,3.1,3.5,3.8,3.9,4.5,6]
plt.hist(values, bins=3)
plt.show()
```



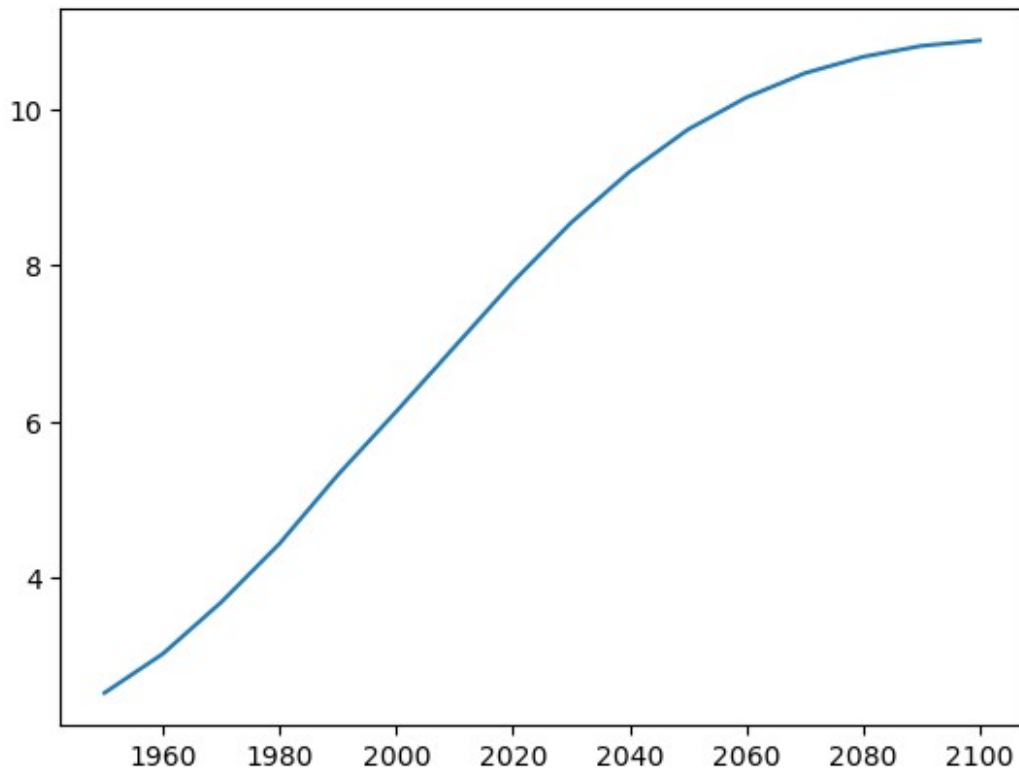
```
import matplotlib.pyplot as plt
year = [1950, 1970, 1990, 2010, 2030, 2050, 2075, 2100]
pop = [2.53, 3.69, 5.26, 6.97, 8.54, 9.73, 10.5, 10.85]

plt.plot(year , pop )
plt.show()
```

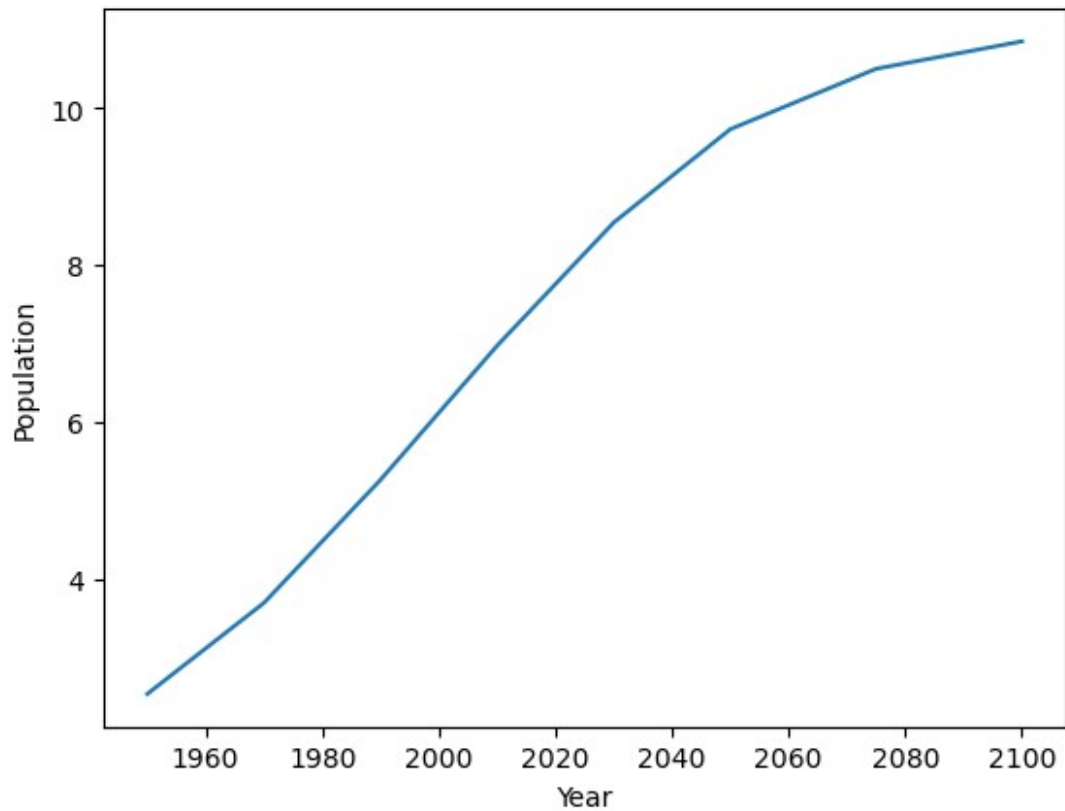


```
#exemple4_Basic plot
import matplotlib.pyplot as plt
year = [1950, 1960, 1970, 1980, 1990, 2000, 2010, 2020, 2030, 2040,
2050, 2060, 2070, 2080, 2090, 2100]
pop = [2.53, 3.03, 3.69, 4.44, 5.32, 6.13, 6.96, 7.79, 8.55, 9.20,
9.74, 10.15, 10.46, 10.67, 10.81, 10.88]
plt.plot(year , pop )

plt.show()
```



```
import matplotlib.pyplot as plt
year = [1950, 1970, 1990, 2010, 2030, 2050, 2075, 2100]
pop = [2.53, 3.69, 5.26, 6.97, 8.54, 9.73, 10.5, 10.85]
plt.plot(year, pop)
plt.xlabel('Year')
plt.ylabel('Population')
plt.show()
```

```
#exemple5_Axis labels
```

```
import matplotlib.pyplot as plt
```

```
year = [1950, 1960, 1970, 1980, 1990, 2000, 2010, 2020, 2030, 2040,  
2050, 2060, 2070, 2080, 2090, 2100]
```

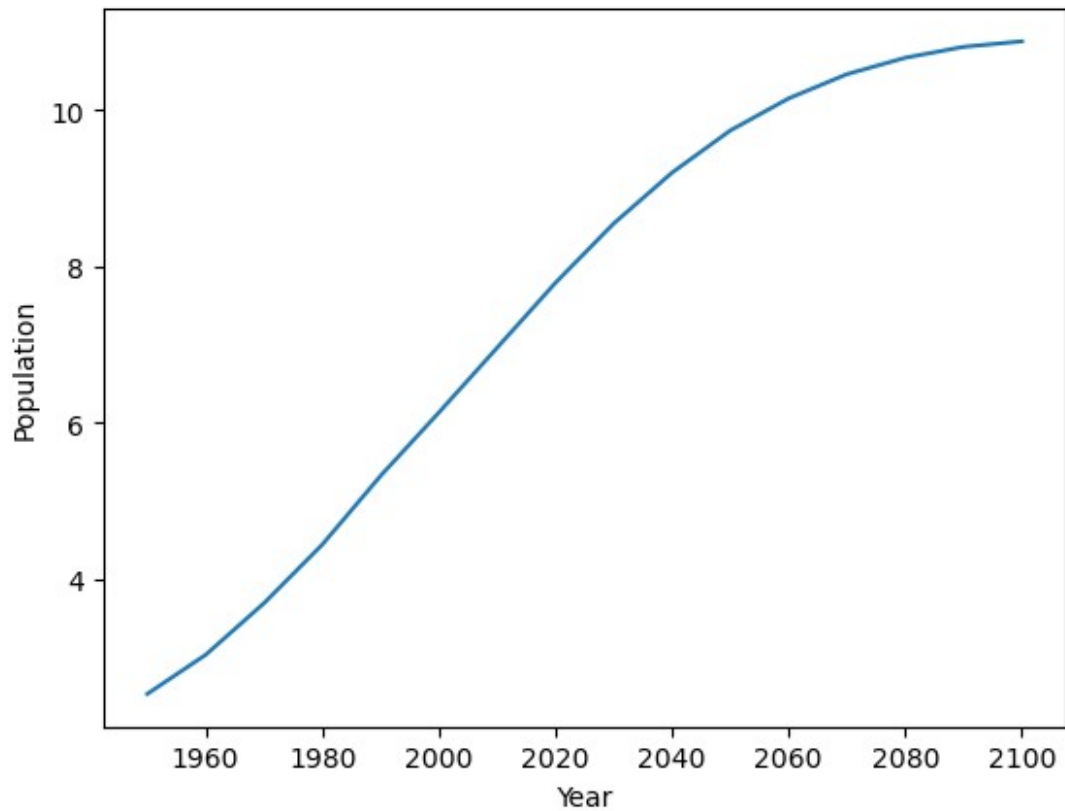
```
pop = [2.53, 3.03, 3.69, 4.44, 5.32, 6.13, 6.96, 7.79, 8.55, 9.20,  
9.74, 10.15, 10.46, 10.67, 10.81, 10.88]
```

```
plt.plot(year, pop)
```

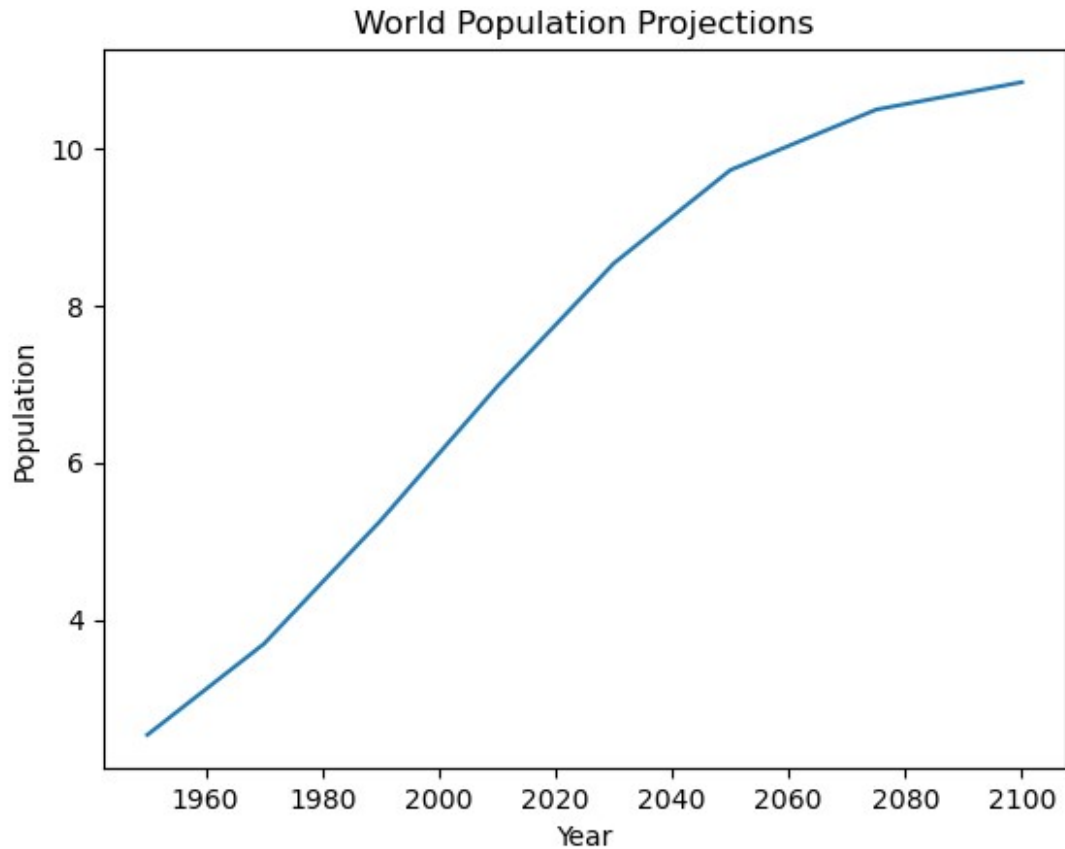
```
plt.xlabel('Year')
```

```
plt.ylabel('Population')
```

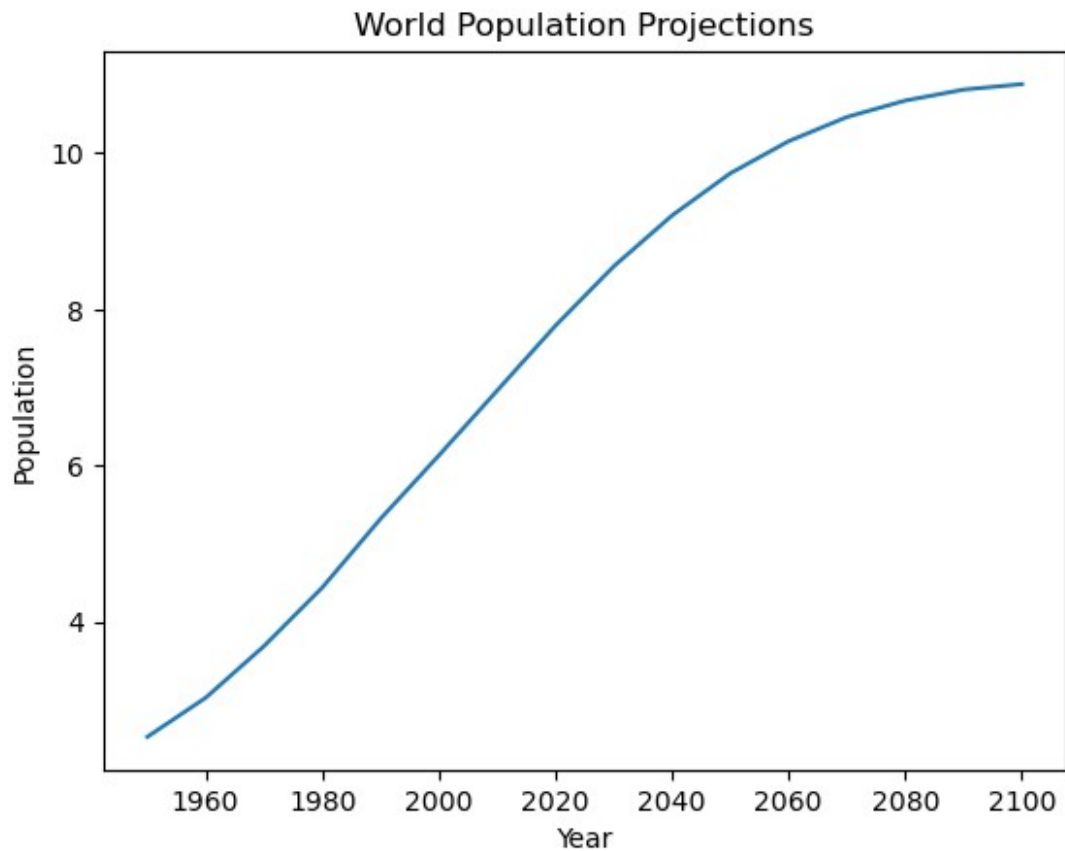
```
plt.show()
```



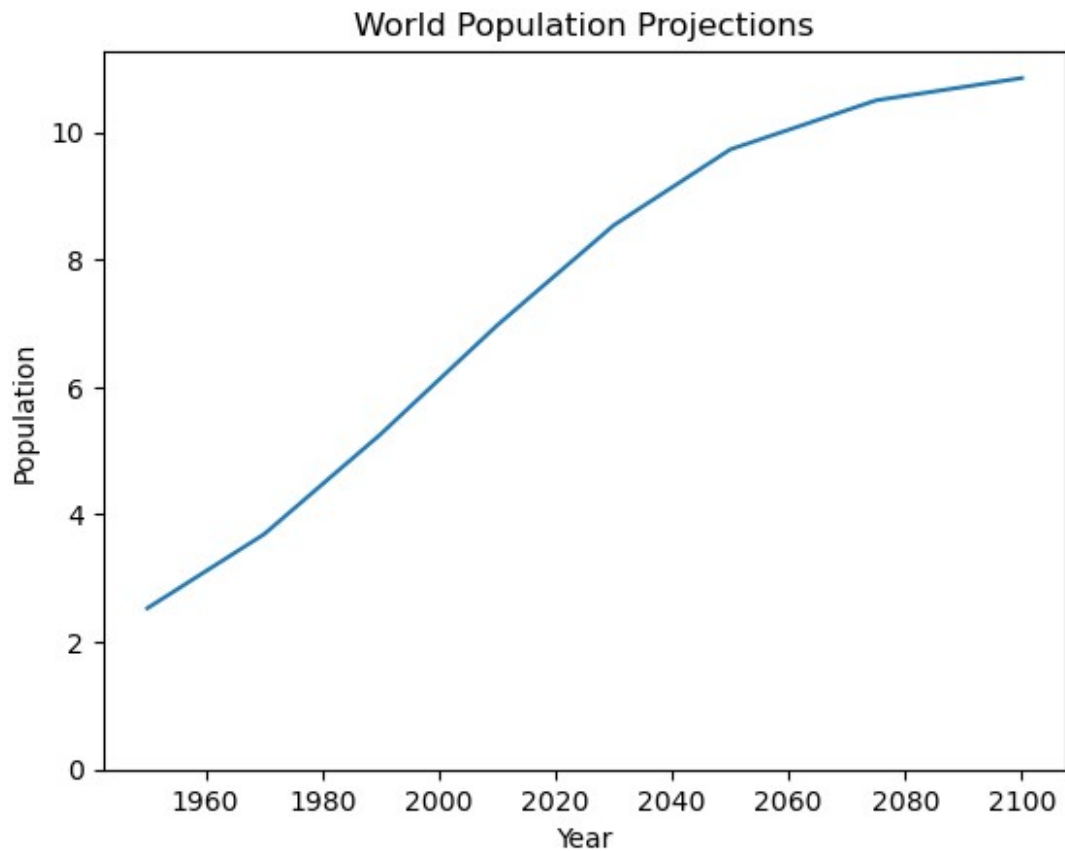
```
import matplotlib.pyplot as plt
year = [1950, 1970, 1990, 2010, 2030, 2050, 2075, 2100]
pop = [2.53, 3.69, 5.26, 6.97, 8.54, 9.73, 10.5, 10.85]
plt.plot(year, pop)
plt.xlabel('Year')
plt.ylabel('Population')
plt.title('World Population Projections')
plt.show()
```



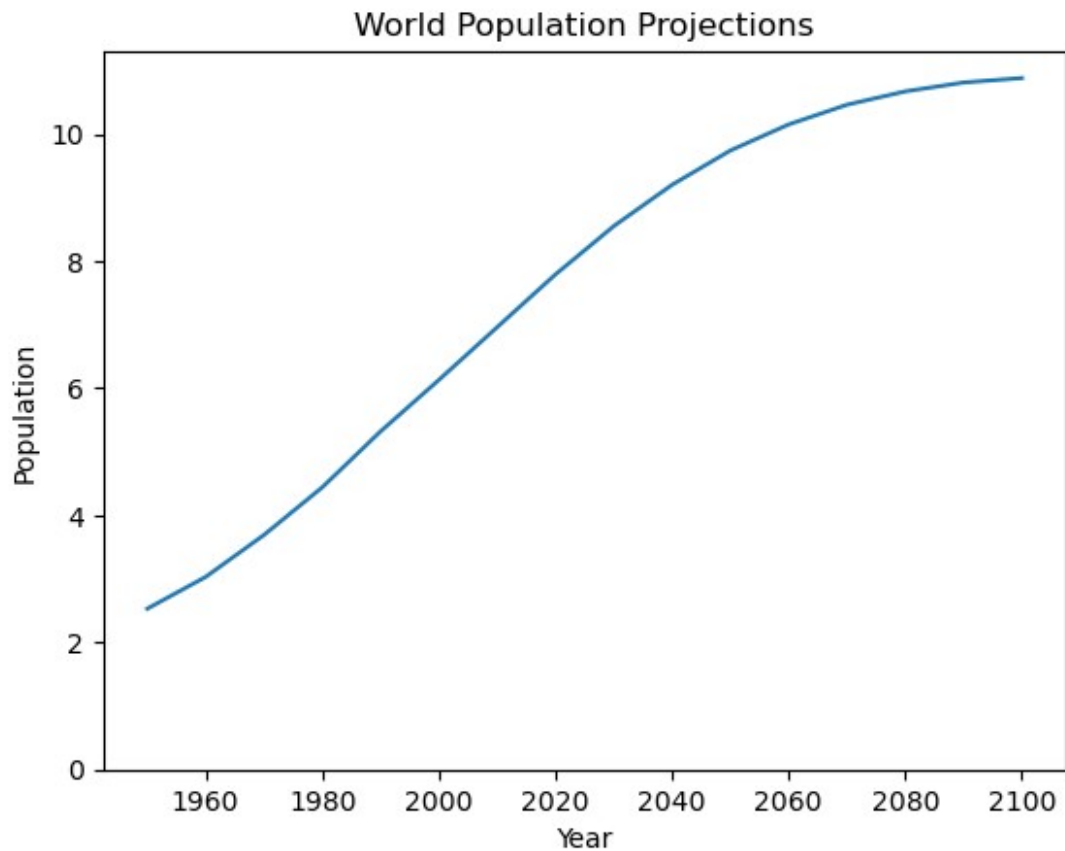
```
#exemple6_Title
import matplotlib.pyplot as plt
year = [1950, 1960, 1970, 1980, 1990, 2000, 2010, 2020, 2030, 2040,
2050, 2060, 2070, 2080, 2090, 2100]
pop = [2.53, 3.03, 3.69, 4.44, 5.32, 6.13, 6.96, 7.79, 8.55, 9.20,
9.74, 10.15, 10.46, 10.67, 10.81, 10.88]
plt.plot(year, pop)
plt.xlabel('Year')
plt.ylabel('Population')
plt.title('World Population Projections')
plt.show()
```



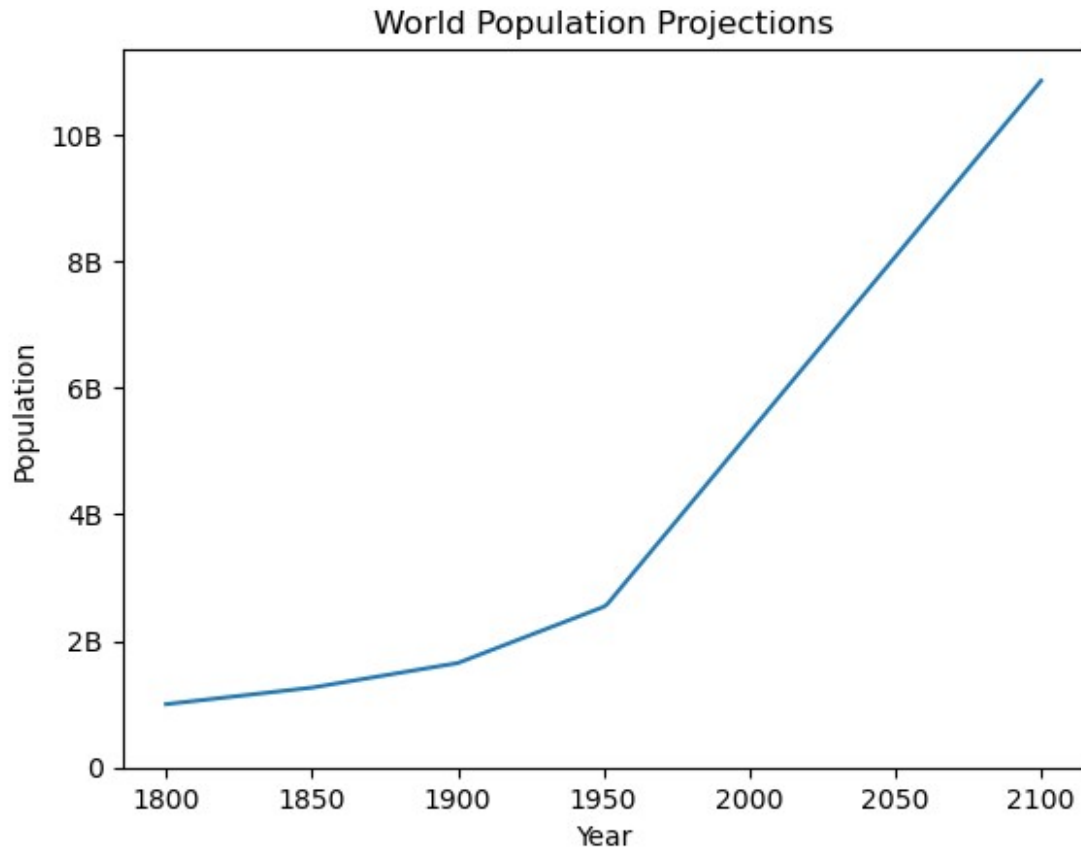
```
import matplotlib.pyplot as plt
year = [1950, 1970, 1990, 2010, 2030, 2050, 2075, 2100]
pop = [2.53, 3.69, 5.26, 6.97, 8.54, 9.73, 10.5, 10.85]
plt.plot(year, pop)
plt.xlabel('Year')
plt.ylabel('Population')
plt.title('World Population Projections')
plt.yticks([0, 2, 4, 6, 8, 10])
plt.show()
```



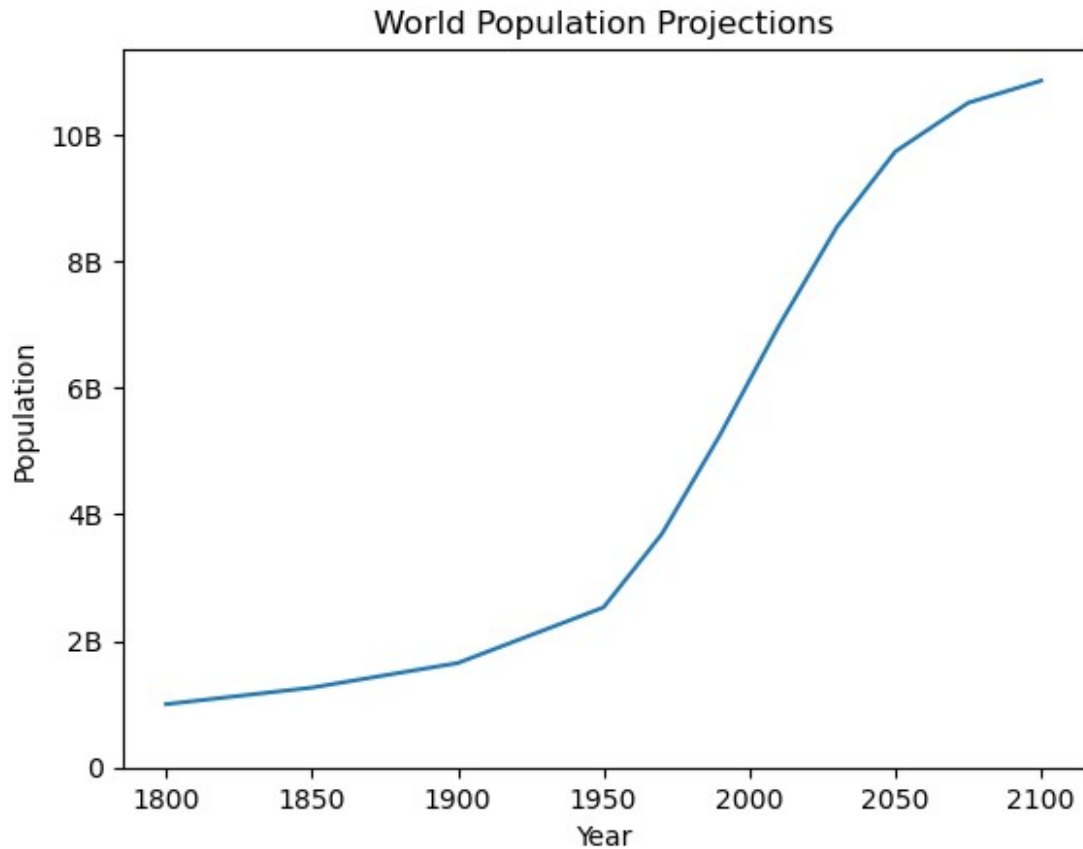
```
#exemple7_Ticks
import matplotlib.pyplot as plt
year = [1950, 1960, 1970, 1980, 1990, 2000, 2010, 2020, 2030, 2040,
2050, 2060, 2070, 2080, 2090, 2100]
pop = [2.53, 3.03, 3.69, 4.44, 5.32, 6.13, 6.96, 7.79, 8.55, 9.20,
9.74, 10.15, 10.46, 10.67, 10.81, 10.88]
plt.plot(year, pop)
plt.xlabel('Year')
plt.ylabel('Population')
plt.title('World Population Projections')
plt.yticks([0, 2, 4, 6, 8, 10])
plt.show()
```



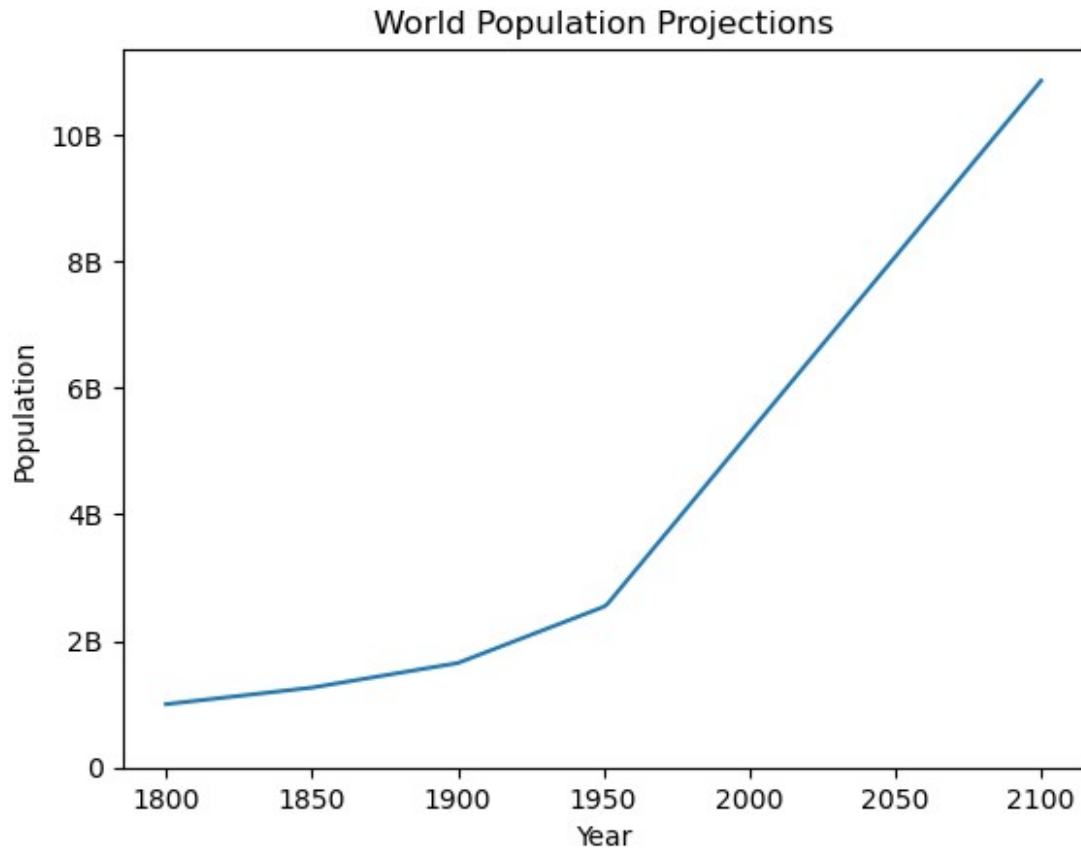
```
import matplotlib.pyplot as plt
year = [1950, 1951, 1952, 2100]
pop = [2.538, 2.57, 2.62, 10.85]
# Add more data
year = [1800, 1850, 1900] + year
pop = [1.0, 1.262, 1.650] + pop
plt.plot(year, pop)
plt.xlabel('Year')
plt.ylabel('Population')
plt.title('World Population Projections')
plt.yticks([0, 2, 4, 6, 8, 10],
['0', '2B', '4B', '6B', '8B', '10B'])
plt.show()
```



```
#exemple8_Ticks (2)
import matplotlib.pyplot as plt
year = [1950, 1970, 1990, 2010, 2030, 2050, 2075, 2100]
pop = [2.53, 3.69, 5.26, 6.97, 8.54, 9.73, 10.5, 10.85]
# Add more data
year = [1800, 1850, 1900] + year
pop = [1.0, 1.262, 1.650] + pop
plt.plot(year, pop)
plt.xlabel('Year')
plt.ylabel('Population')
plt.title('World Population Projections')
plt.yticks([0, 2, 4, 6, 8, 10],
['0', '2B', '4B', '6B', '8B', '10B'])
plt.show()
```



```
import matplotlib.pyplot as plt
year = [1950, 1951, 1952, 2100]
pop = [2.538, 2.57, 2.62, 10.85]
# Add more data
year = [1800, 1850, 1900] + year
pop = [1.0, 1.262, 1.650] + pop
plt.plot(year, pop)
plt.xlabel('Year')
plt.ylabel('Population')
plt.title('World Population Projections')
plt.yticks([0, 2, 4, 6, 8, 10],
['0', '2B', '4B', '6B', '8B', '10B'])
plt.show()
```

```
#exemple9_historical data
import matplotlib.pyplot as plt
year = [1950, 1970, 1990, 2010, 2030, 2050, 2075, 2100]
pop = [2.53, 3.69, 5.26, 6.97, 8.54, 9.73, 10.5, 10.85]
# Add more data
year = [1800, 1850, 1900] + year
pop = [1.0, 1.262, 1.650] + pop
plt.plot(year, pop)
plt.xlabel('Year')
plt.ylabel('Population')
plt.title('World Population Projections')
plt.yticks([0, 2, 4, 6, 8, 10],
            ['0', '2B', '4B', '6B', '8B', '10B'])
plt.show()
```

